"Dissemination of Education for Knowledge, Science and Culture"

-Shikshanmaharshi Dr. Bapuji Salunkhe

VIVEKANAND COLLEGE, KOLHAPUR

(Empowered Autonomous)

DEPARTMENT OF STATISTICS

A Project Report on

"A Study on Student Challenges and Their Impact on Academic Achievement."

Submitted by –

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in partial fulfilment for the award of the degree of

MASTER OF SCIENCE

in

STATISTICS

DEPARTMENT OF STATISTICS

VIVEKANAND COLLEGE

KOLHAPUR

2025-26

CERTIFICATE

This is to Certify that,

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Have satisfactorily completed the research project work on **"A Study on Student Challenges and Their Impact on Academic Achievement"** as part of Research Project course for **M.Sc. II**, prescribed by the Department of Statistics, **Vivekanand College, Kolhapur (An Empowered Autonomous Institute)** in the academic year **2025-26**.

This research project has been completed under our guidance and supervision. To the best of our knowledge and belief, the matter presented in this project report is original and has not been submitted elsewhere for any other purpose.

Date: 29/04/2026

Place: Kolhapur

Project Guide | **Examiner** | **Head of Department**

(Mr. A. B. Bhosale) | | (Mrs. V. C. Shinde)

Acknowledgment

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Abstract

Academic performance is a key indicator of student success and the overall quality of education. This study focuses on understanding how academic, personal, and environmental factors influence students' academic performance, along with identifying the major challenges faced by them. The study is based on primary data collected from undergraduate students through a structured questionnaire.

The findings indicate that academic performance is not determined by a single factor but is influenced by a combination of multiple aspects. Academic-related factors such as attendance and internal performance play an important role, while personal and environmental factors including stress, family support, and study habits also have a significant influence.

The study further highlights common challenges faced by students, such as lack of motivation, difficulty in understanding subjects, and problems related to time management. Overall, the results emphasize the importance of a balanced and comprehensive approach that considers both academic and non-academic aspects to improve student performance. The insights from this study can be useful for educators and policymakers in developing effective strategies to support students.

Motivation

The motivation behind this study arises from the growing concern about students' academic performance and the various challenges they face during their educational journey. In today's competitive academic environment, students are not only expected to perform well in examinations but also manage multiple personal, social, and environmental pressures.

Many students experience difficulties such as lack of motivation, stress, time management issues, and problems in understanding subjects, which directly or indirectly affect their academic achievement. However, these challenges are often overlooked, and academic performance is usually judged only on the basis of marks and grades.

This study is motivated by the need to go beyond traditional evaluation methods and to understand the real factors influencing student performance. By identifying and analysing both academic and non-academic factors such as study habits, family support, stress levels, and learning difficulties, the study aims to provide a more comprehensive understanding of student challenges.

Another important motivation is to use statistical and data-driven approaches to examine these factors scientifically. This helps in identifying patterns, relationships, and key problem areas that may not be visible through simple observation.

Ultimately, the study is motivated by the goal of helping students improve their academic performance by identifying their difficulties and suggesting areas where support is needed. The findings of this research can assist educators, institutions, and policymakers in developing effective strategies to support students and enhance overall educational outcomes.

Introduction

Academic performance is one of the most important indicators of students' success and the quality of education provided by institutions. In recent years, the education sector has become more competitive, and institutions are increasingly focusing on improving student outcomes and reducing academic failure. However, student performance is a complex phenomenon influenced by multiple factors, including academic behaviour, personal characteristics, and environmental conditions.

With the rapid growth of educational data, there has been a significant shift towards data-based analysis in education. Large amounts of student data are generated through academic activities such as attendance, assignments, tests, and examinations. Analysing this data helps in understanding student learning patterns and identifying factors that affect academic achievement. In this context, modern analytical approaches have made it possible to extract useful information from student data and support better decision-making in education.

Previous studies have shown that student performance can be analysed and predicted using different types of variables. Academic factors such as attendance, internal assessment scores, and previous performance are considered important indicators of student achievement. At the same time, personal and behavioural factors such as study habits, motivation, stress, and time management also influence performance. In addition, environmental factors such as family background, income level, and access to resources play a supportive role in shaping students' academic outcomes.

Many researchers have applied various analytical and data mining techniques to study student performance. These techniques help in identifying patterns in data and predicting outcomes such as pass or fail, grades, and academic progress. Such analysis is useful for identifying students who may be at risk and for providing timely academic support. However, most of these studies focus mainly on prediction using limited academic data, which may not fully capture the real challenges faced by students.

Despite the progress in this area, there is still a need to analyse student performance from a broader perspective. Academic performance should not be studied only through marks and grades, but also through the challenges and conditions that affect students' learning. Factors such as lack of motivation, difficulty in understanding subjects, family-related issues, and lack of proper study environment can significantly impact performance. Therefore, a comprehensive approach that considers multiple dimensions is necessary.

The present study is conducted to analyse the impact of academic, personal, and environmental factors on students' academic performance. It also aims to identify the key challenges faced by students and examine their relationship with academic achievement. By using statistical and analytical techniques, the study provides a better understanding of student performance and helps in identifying areas where support is required.

The findings of this study will be useful for educators, institutions, and policymakers in designing effective strategies to improve student performance and provide necessary support to students facing difficulties in their academic journey.

Review of Literature

Several researchers have studied the factors that affect student performance and how to identify students who may be at risk of failing.

Ramaswami and Bhaskaran (2009**)** investigated which student attributes (like marks and attendance) are most important for performance prediction. Their study emphasized the value of selecting the right data for better results. They showed that using fewer but important variables improves prediction accuracy. Their findings also suggest that proper feature selection helps in reducing complexity and improving model efficiency.

Yadav et al. (2012) focused on using academic information like attendance and test scores to group students based on performance. They highlighted how early identification can reduce failure rates. The study also classified students into different performance categories such as high, average, and low performers. This helps teachers take early action to support weak students.

Vamanan and Ramesh (2013) studied how students' personal information, family background, and school environment impact academic performance. They found that past grades, parents' occupation, and private tuition strongly affect success. The study also showed that combining personal and academic data gives better prediction results. It highlights the importance of considering multiple factors together.

Adhatrao et al. (2013) analyzed student background and academic data to develop a prediction system. They aimed to help institutions identify students needing academic support. Their study also demonstrated that prediction systems can be implemented practically using simple tools. It helps institutions take preventive measures before students fail.

Sujatha et al. (2018) studied first-year engineering students to forecast academic performance based on earlier records. The research aimed to support early intervention for struggling students. The study also highlighted that early prediction can improve student outcomes. It emphasizes the importance of continuous monitoring of student performance.

Li et al. (2019) grouped students by analyzing academic scores in various subjects. Their study aimed to help teachers better understand learning outcomes and provide more targeted support. It also showed that students may perform differently across subjects. This helps in identifying specific areas where students need improvement.

Zhang, Yupei et al. (2021) examined academic records of over 1800 university students to predict final grades. The study showed that basic academic data can effectively identify at-risk students early. It also proved that large datasets can improve prediction reliability. The study highlights the usefulness of simple academic records in decision-making.

Lynn and Emanuel (2021) reviewed past research from 2010 to 2020 on student performance prediction. They discussed common techniques and encouraged choosing suitable models based on context. The study also pointed out that no single method works best in all situations. It suggests that model selection should depend on the type of data and objective.

Ahmad et al. (2021) reviewed 78 papers to understand how machine learning is applied to predict student outcomes. They found many models lack follow-up actions for student support after prediction. The study also highlighted the need for practical implementation of prediction systems. It suggests combining prediction with intervention strategies.

Mustapha (2023) explored how academic data like assignments and course details can be used to identify students at risk. The study emphasized early detection and better decision-making by educators. It also showed that selecting important features improves model performance. The research supports data-driven approaches in education.

Hussain et al. (2024) worked on predicting final exam results using regular academic data. Their study focused on helping teachers identify weak students early for better support. It also showed that accurate prediction models can improve teaching strategies. The study highlights the role of modern techniques in improving academic outcomes.

Objectives of the Study

1. To identify and test the influence of various academic, personal, and environmental factors on students' academic performance.
2. To identify student difficulties in their studies using their information.
3. To examine and test the relationship between students' personal experiences, study habits, and their academic achievement.
4. To identify at-risk students, analyse student challenges, and study their support needs.

Data Collection and Research Methodology

Data Collection

This study aims to examine student challenges and their impact on academic performance in Kolhapur District. The research follows a quantitative cross-sectional survey design to analyse how academic, personal, and environmental factors influence undergraduate students' academic achievement.

The target population consists of undergraduate students from various degree colleges across urban and rural areas of Kolhapur District, including Arts, Science, Commerce, and Management streams. To ensure proper representation, a multistage stratified sampling technique with proportional allocation was used. Initially, the district was divided into urban and rural areas. Further stratification was done based on academic streams, and institutions within each stratum were treated as clusters. Colleges were selected using simple random sampling, and students were randomly chosen from the selected institutions.

The dataset used in this study is based on primary data collected through a structured questionnaire designed for academic research purposes. The dataset, titled *"Student Academic Performance and Challenges Dataset (2025–26)"*, consists of 283 observations and approximately 35–40 variables. Each row represents an individual student respondent, while each column corresponds to a specific variable obtained from the questionnaire. The data is maintained in Excel/CSV format and includes information related to academic performance, study habits, personal factors, and challenges faced by students.

Description of Variables:

Variable Name	Description
Gender	Gender of the student
Area	Residential area of the student (Rural/Urban)
Faculty	Academic stream (Arts, Science, Commerce, Management)
College_Location	Location of the college (Rural/Urban)
Father_Education	Educational qualification of father
Mother_Education	Educational qualification of mother
Father_Occupation	Occupation of father
Mother_Occupation	Occupation of mother
Family_Income	Annual family income category
Attendance	Average class attendance percentage
Lecture_Attendance	Frequency of attending lectures
Assignment_Submission	Regularity in submitting assignments
Teacher_Help	Frequency of academic help from teachers
Study_Hours	Daily study hours
Time_Management	Ability to manage study and other activities
Study_Methods	Methods used for studying (multiple responses)
Previous_Percentage	Academic performance in previous year (%)
Internal_Performance	Self-evaluation of internal assessment performance
Backlogs	Number of backlogs in previous semester
Internet_Access	Availability of internet for study
Online_Usage	Frequency of using online platforms for study
Other_Responsibilities	Impact of other responsibilities on study time
Motivation_Level	Level of study motivation
Family_Problems	Presence of family-related issues
Concentration_Level	Ability to concentrate while studying
Health_Issues	Health-related challenges
Understanding_Difficulty	Difficulty in understanding subjects
Study_Time_Lack	Lack of sufficient study time
Distractions	Factors causing distraction during study
Parental_Support	Support from parents for education
Motivation_Factors	Factors motivating students to study
Stress_Level	Level of stress or anxiety related to studies
Family_Pressure	Academic pressure from family
Sleep_Hours	Average daily sleep hours
Support_Needed	Type of support required for better study

Flowchart of Research Methodology

The following flowchart represents the step-by-step procedure adopted for conducting the research study.

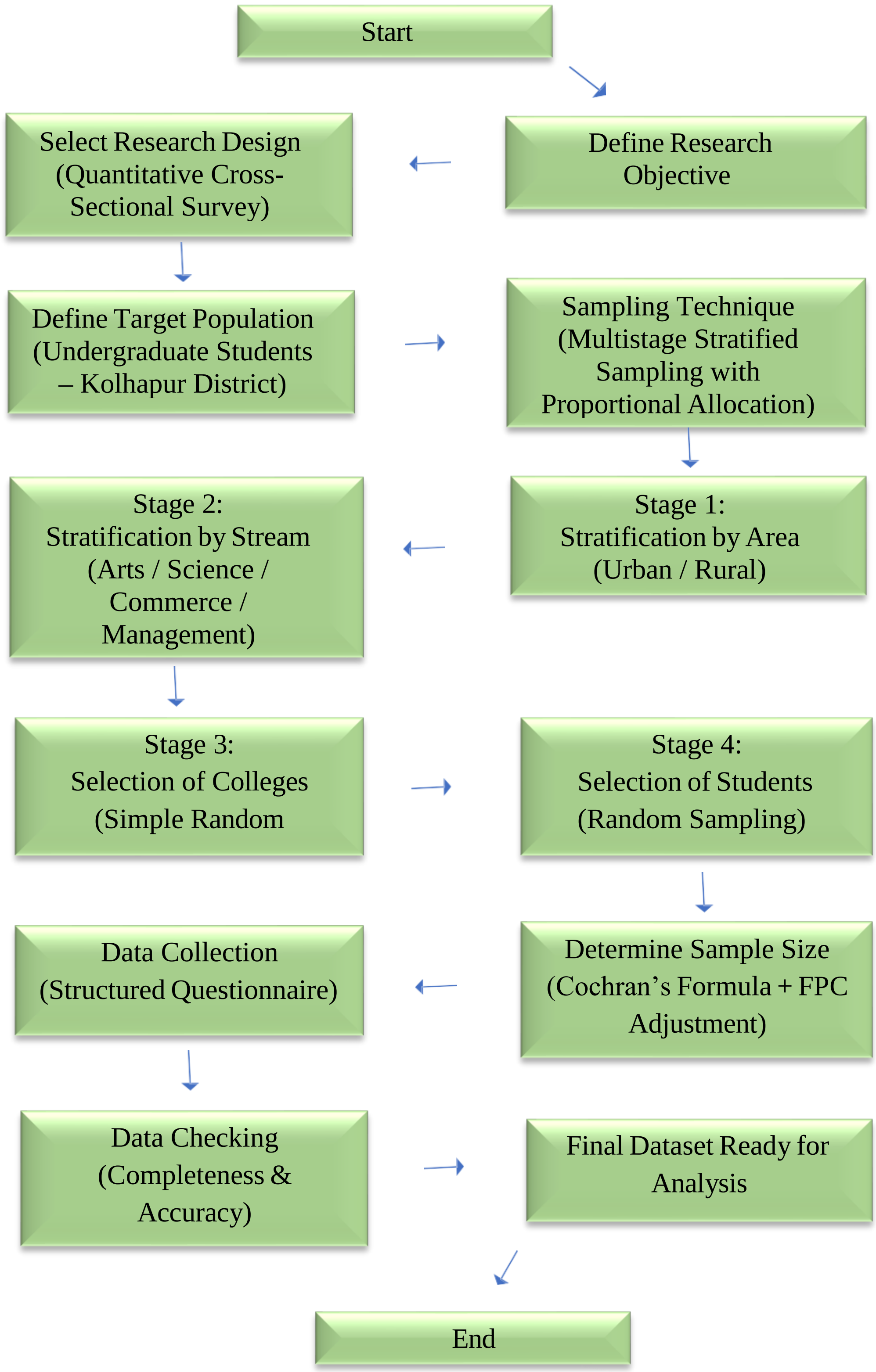


Fig: flowchart Diagram

Statistical Software

1. Python:

Python was used as the primary software for data cleaning, preprocessing, exploratory data analysis, statistical analysis, visualization, and the development of machine learning models such as Logistic Regression, Random Forest, and XGBoost.

2.MS-Excel:

MS-Excel was used for initial data entry, data verification, coding of questionnaire responses, and basic tabulation.

3.JupyterNotebook:

Jupyter Notebook served as the development environment for executing Python code, performing analysis, visualizing results, and documenting the complete research workflow.

Objective 1

To identify and test the influence of various academic, personal, and environmental factors on students’ academic performance

- The analysis was carried out using descriptive statistics, t-test, ANOVA, chi-square test, correlation analysis, and multiple linear regression.

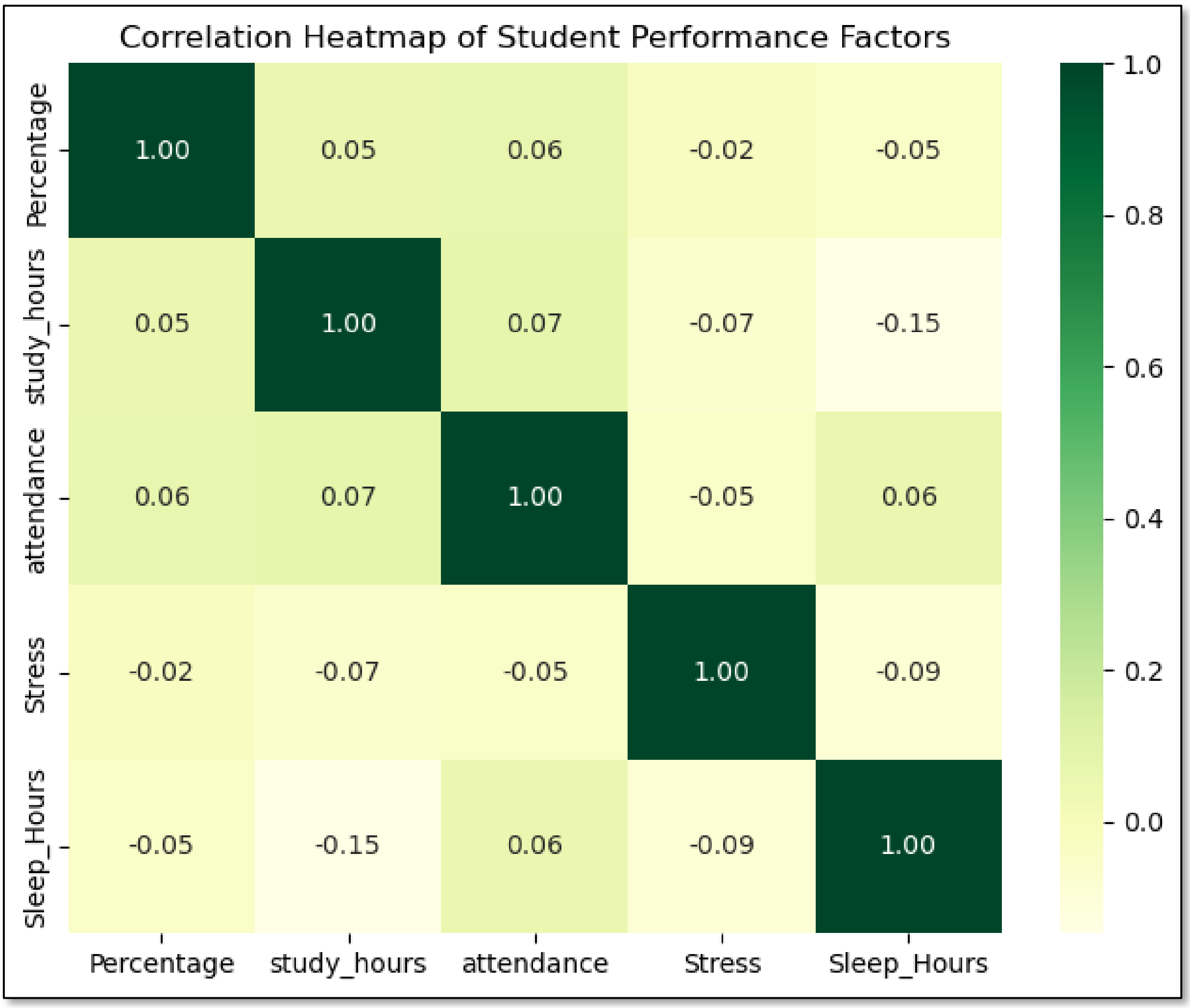
➢ **Descriptive Statistical Analysis of Key Variables:**

	Percentage	study_hours	attendance	Stress	Sleep_Hours
count	283	283	283	283	**283**
mean	72.5833	2.5865	2.4310	2.4558	**2.7385**
std	10.4549	0.8392	0.7229	0.8125	**0.7594**
min	35	1	1	1	**1**
25%	65	2	2	2	**2**
50%	74.24	3	3	2	**3**
75%	79	3	3	3	**3**
max	**95**	**4**	**3**	**4**	**4**

Interpretations:

Students have moderate academic performance with some variation, and factors like study hours, attendance, stress, and sleep are mostly at average levels. This shows performance is not same for all students.

➢ **Correlation Analysis of Academic Performance and Study Variables:**



Interpretations:

Academic performance shows very weak correlation with all factors, indicating that no single variable has a strong influence; it is affected by a combination of multiple factors.

- **Multiple Linear Regression Analysis of Factors Affecting Academic Performance:**

Table 4.X: Multiple Linear Regression Coefficients

Variable	Coefficient (β)	Std. Error	t-value	p-value
Constant	53.9066	6.95	7.756	**0**
Study Hours	-0.0814	0.77	-0.106	**0.916**
Attendance	0.0109	0.894	0.012	**0.99**
Assignment Submission	0.4721	1.022	0.462	**0.645**
Teacher Support	1.0192	0.706	1.444	**0.15**
Motivation Score	0.1908	0.156	1.221	**0.223**
Stress	-0.2143	0.771	-0.278	**0.781**
Sleep Hours	-0.3885	0.816	-0.476	**0.635**
Study Management	-0.5059	0.87	-0.581	**0.561**
Family Income	0.343	0.787	0.436	**0.663**
Internet Access	0.4621	0.588	0.785	**0.433**
Parent Support	7.3394	4.677	1.569	**0.118**
Internal Performance	**2.748**	**0.627**	**4.386**	**0**

Table 4.Y: Model Summary of Regression Analysis

Statistic	Value
R-squared	**0.122**
Adjusted R-squared	**0.083**
F-statistic	**3.115**
Prob (F-statistic)	**0.000369**
Number of Observations	**283**

Interpretations:

Multiple Linear Regression analysis was conducted to identify factors affecting academic performance.

a) The overall model is statistically significant ($p = 0.000369$), but the R^2 value (0.122) indicates that only 12.2% of the variation is explained, suggesting other factors also influence performance.
b) Internal performance is the only significant predictor ($\beta = 2.748$, $p < 0.05$), showing a strong positive effect on academic performance.
c) Other variables such as study hours, attendance, stress, sleep hours, motivation, and income are not statistically significant.

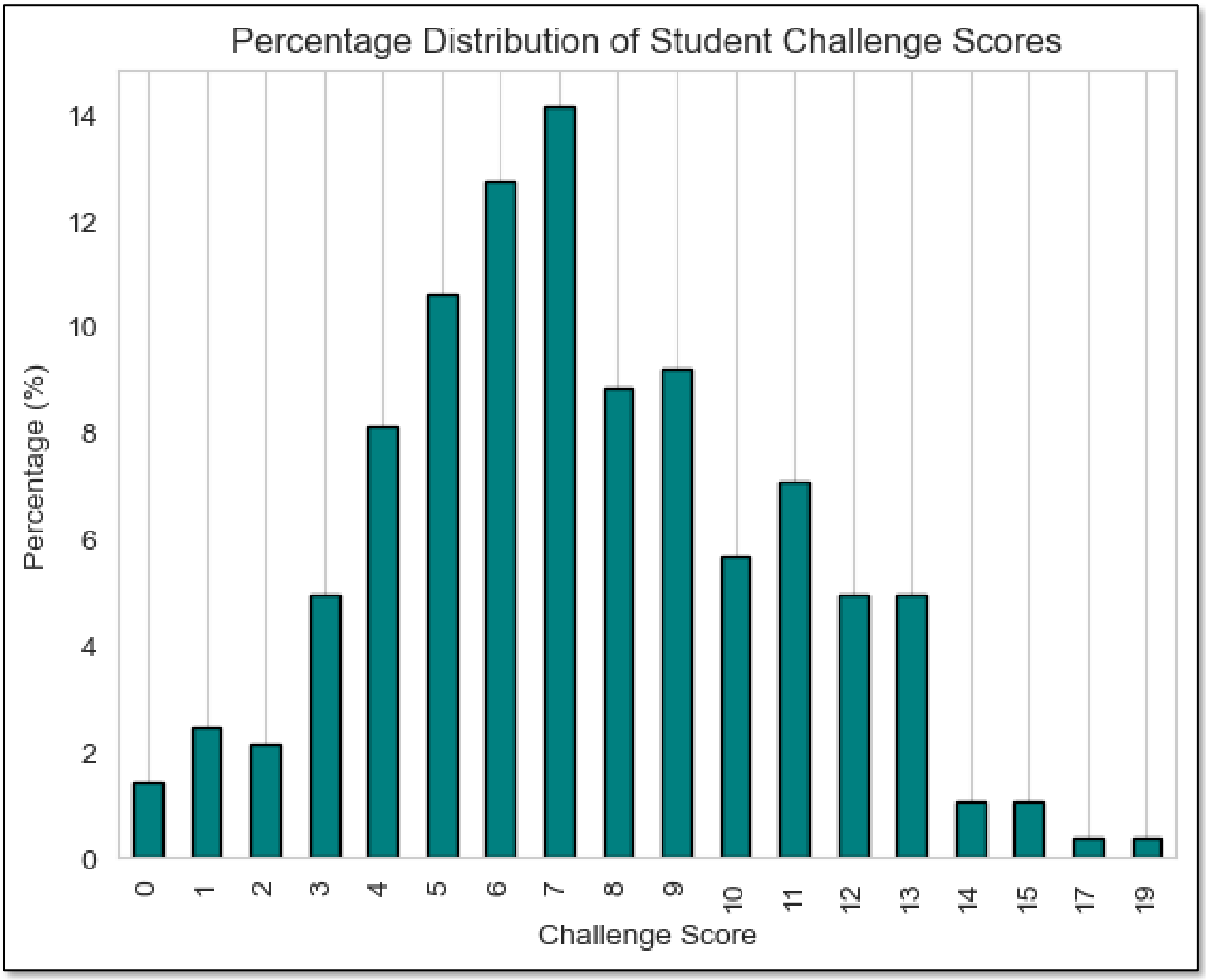
Internal performance is the most important factor, while other variables have a relatively weak impact.

Objective 2

To identify student difficulties in their studies using their information

- The analysis was carried out using descriptive statistics, frequency distribution, Likert scale analysis, reliability analysis (Cronbach's Alpha), factor analysis, and composite challenge index analysis.

➢ **Frequency and Percentage Analysis of Student Challenges:**



Interpretations:

a) Frequency and percentage distribution of challenge scores were calculated.

b) The majority of students fall in the moderate challenge score range.

c) A smaller number of students experience low levels of challenges.

d) Very few students face high levels of challenges.

Overall, it indicates that most students experience a moderate level of academic and personal challenges.

Challenge Score is calculated by assigning numerical values to Likert-scale responses for different student challenges and then summing these values to form a composite index representing the overall level of challenges faced by each student.

➢ Comparative Analysis of Challenge Score Across Student Groups:

Mean Challenge Score -

Mean Challenge Score: **7.3710**

Figure X: Mean Challenge Score across Academic Streams -

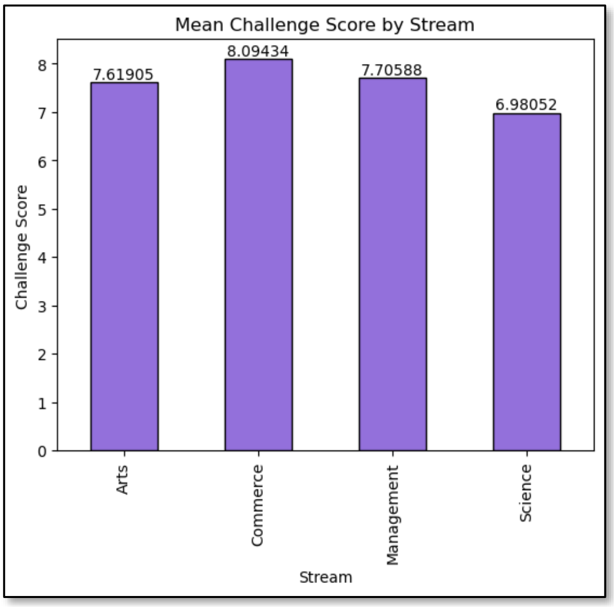


Figure X: Mean Challenge Score across Area (Urban vs Rural) -

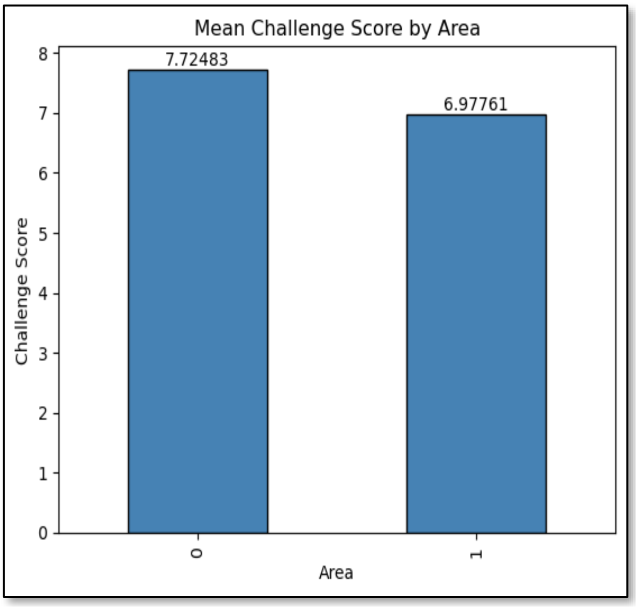
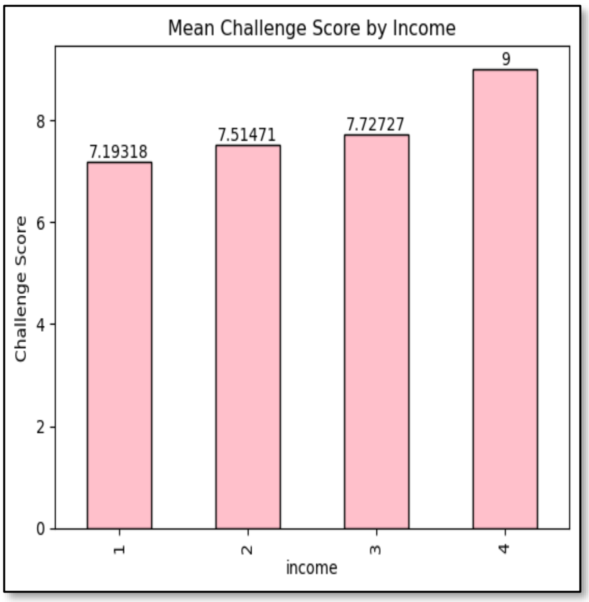


Figure X: Mean Challenge Score across Income Groups -



Interpretations:

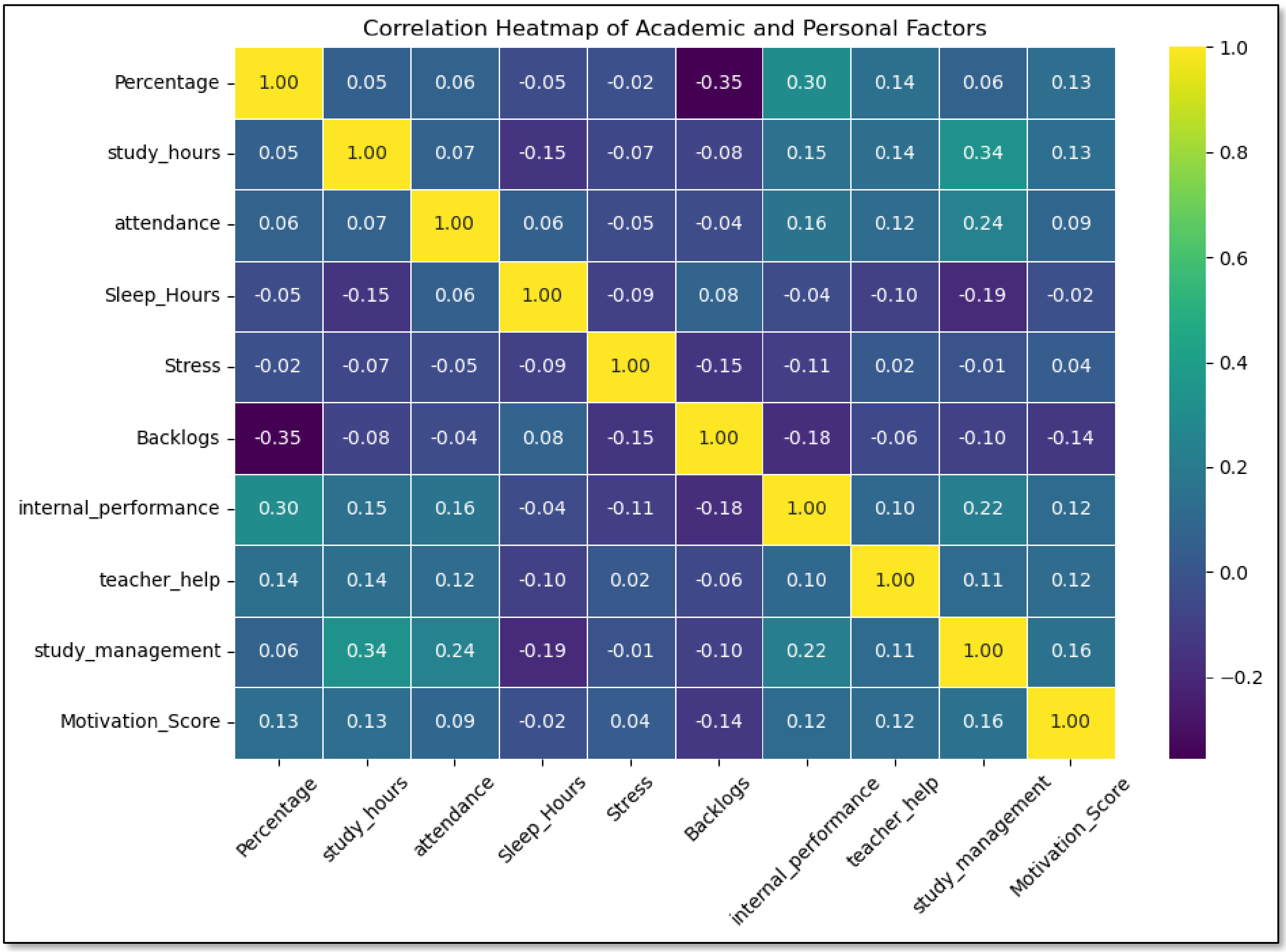
a) Overall students face moderate level of challenges.
b) Commerce students and rural students face more challenges compared to others.
c) This shows student challenges are different across groups and not the same for all students.

Objective 3

To examine and test the relationship between students' personal experiences, study habits, and their academic achievement.

- The analysis was carried out using correlation analysis, chi-square test of independence, independent sample t-test, and one-way ANOVA.

➢ Correlation Analysis of Factors Affecting Students' Academic Performance:



Interpretations:

• Internal Performance, Study Management, and Teacher Help show a positive association with academic performance.
• Backlogs show the strongest negative association with academic performance.
• Stress has a very weak negative relationship with academic performance.

• Overall, Internal Performance positively influences academic achievement, while Backlogs adversely affect student performance. Most other variables exhibit only weak correlations.

➢ Chi-square Test of Independence for Categorical Variables:

To examine the association between backlogs and internet access, the Chi-square test of independence was applied. This test helps to determine whether two categorical variables are related or independent.

Chi-square Test of Independence between Backlogs and Internet Access –

Data Preparation:

To satisfy the assumptions of the Chi-square test, categories were combined as follows:

- Backlogs were grouped into:

- 0 → No Backlogs
- 1, 2 → Has Backlogs

- Internet Access was grouped into:

- 0, 1 → Low Access
- 2, 3, 4 → High Access

Hypothesis:

- Null Hypothesis (H_0): There is no association between Backlogs and Internet Access.
- Alternative Hypothesis (H_1): There is an association between Backlogs and Internet Access.

- Contingency table:

Backlogs / internet_access	High	Low
0	218	**16**
1	**44**	**5**

Results:

Chi-square value: 0.2681

P-value: 0.6045

-Since the p-value is greater than 0.05, the null hypothesis is not rejected.

-There is no significant association between backlogs and internet access.

Interpretations:

The Chi-square test was conducted to examine the association between backlogs and internet access.

Internet access does not have a significant impact on the presence of backlogs among students.

➢ Chi-square Test of Independence between Attendance and Area :

Hypothesis:

- Null Hypothesis (H_0): There is no association between attendance and area.
- Alternative Hypothesis (H_1): There is an association between attendance and area.

- Contingency table:

Attendance/ Area	0	1
1	18	**21**
2	50	**33**
3	**81**	**80**

Results:

Chi-square value: 2.9320
P-value: 0.2308

-The p-value (0.231) is greater than the significance level of 0.05.

-Therefore, the null hypothesis is not rejected.

Interpretations:

The Chi-square test was conducted to examine the association between attendance and area.

This indicates that there is no statistically significant association between attendance and area.

Attendance levels are similar among students from both urban and rural areas.

➢ Chi-square Test of Independence between Family Income and Academic Performance Category:

Data Preparation:

Due to low expected frequencies in some cells, income categories were combined as:

Income Categories:

- 1, 2 → Low Income
- 3, 4 → High Income

Academic Performance Categories:

- Below 75% → Below Average
- 75% and above → Above Average

Hypothesis:

- Null Hypothesis (H_0): There is no association between family income and academic performance category.
- Alternative Hypothesis (H_1): There is an association between family income and academic performance category.

- Contingency table:

Income/Performance	Below Average	Above Average
High	26	**13**
Low	**149**	**95**

Results:

Chi-square value: 0.2411

P-value: 0.6233

- The p-value (0.6233) is greater than 0.05.
- Therefore, the null hypothesis is accepted.

Interpretations:

The Chi-square test was conducted to examine the association between family income and academic performance category.

This indicates that there is no statistically significant association between family income and academic performance.

Family income does not have a significant influence on students’ academic performance.

➢ Independent Sample t-test for Comparison of Academic Performance between Urban and Rural Students:

To compare the academic performance of students from urban and rural areas, an Independent Sample t-test was conducted. This test is used to determine whether there is a statistically significant difference in the mean academic performance between the two groups.

Hypothesis:

- Null Hypothesis (H_0): There is no significant difference in academic performance between Urban and Rural students.
- Alternative Hypothesis (H_1): There is a significant difference in academic performance between Urban and Rural students.

Results:

T-Statistic: 0.9038
P-Value: 0.3668

- The p-value (0.3668) is greater than 0.05.
-Therefore, the null hypothesis is not rejected.

Interpretations:

An Independent Sample t-test was conducted to compare academic performance between Urban and Rural students.

This indicates no statistically significant difference in academic performance between Urban and Rural students.

Area of residence does not significantly influence academic performance in this study.

➢ One-way ANOVA for Comparison of Academic Performance across Different Streams:

To examine whether there is a significant difference in academic performance among students from different streams, a one-way ANOVA test was conducted. This test is used to compare the mean academic performance across more than two groups such as Arts, Science, Commerce, and Management.

Hypothesis:

- Null Hypothesis (H_0): There is no significant difference in academic performance among different streams.
- Alternative Hypothesis (H_1): There is a significant difference in academic performance among different streams.

Results:

F-Statistic: 1.4156
P-Value: 0.2384

-The p-value (0.2384) is greater than 0.05.
-Therefore, the null hypothesis is not rejected.

Interpretations:

One-way ANOVA was conducted to compare academic performance across different streams.

This indicates no statistically significant difference in academic performance among streams.

Students from different streams have similar levels of academic performance.

Objective 4

To identify at-risk students, analyse student challenges, and study their support needs.

- The analysis was carried out using Logistic Regression, Exploratory Factor Analysis (EFA), and Support Needs Analysis (frequency, cross-tabulation, and Chi-square test).

➢ Identification of At-Risk Students using Logistic Regression:

Logistic Regression analysis was applied to identify students who are academically at risk. This method is suitable for binary classification and helps in estimating the probability of a student falling into the "At-Risk" category.

Criteria for At-Risk Classification:

Students were classified as "At-Risk" based on the following conditions:

- Attendance less than 60%
- Previous Percentage less than 50%
- Presence of Backlogs

Binary Variable Definition:

- 0 = Not at Risk
- 1 = At Risk

Model Specification:

- **Dependent Variable:** At-Risk Status (0/1)
- **Independent Variables:** Study Hours, Attendance, Motivation Score, Stress Level, Sleep Hours, Family Income, Internet Access, Parent Support, Internal Performance.

Purpose of the Model: This model helps in identifying the probability of a student being at academic risk and highlights the key factors contributing to student vulnerability, enabling early intervention and support.

Results:

Model Performance Evaluation:

Confusion Matrix:

	Predicted: Not At-Risk (0)	Predicted: At-Risk (1)
Actual: Not At-Risk (0)	46	**1**
Actual: At-Risk (1)	**5**	**5**

Classification Report:

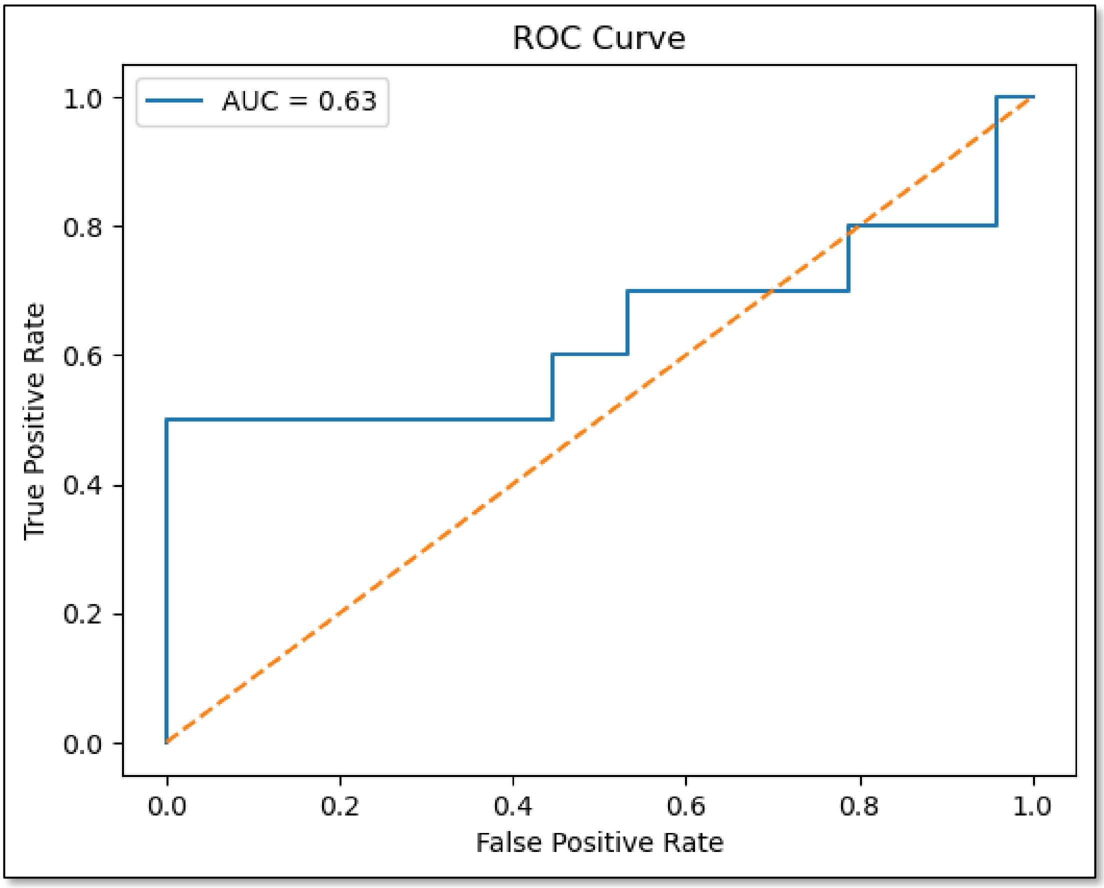
Class	Precision	Recall	F1-Score	Support
Not At-Risk (0)	0.9	0.98	0.94	**47**
At-Risk (1)	**0.83**	**0.5**	**0.62**	**10**

Overall Model Accuracy: 0.8947 (89.47%)

ROC Curve and AUC Analysis:

ROC Curve (Receiver Operating Characteristic Curve):

The ROC curve was plotted to evaluate the performance of the Logistic Regression model in distinguishing between at-risk and not at-risk students.



Area Under the Curve (AUC):

The AUC value obtained for the model is – 0.63

Interpretations:

Logistic Regression was used to identify academically at-risk students.

a) The model achieved an accuracy of 89.47%, indicating good overall performance.
b) Most non-risk students are correctly classified with minimal errors.
c) For at-risk students, recall is 0.50 and precision is 0.83, showing moderate detection ability.
d) The F1-score (0.62) and AUC (0.63) indicate fair model performance.

The model performs well for non-risk students but has moderate effectiveness in identifying at-risk students, suggesting scope for improvement.

➢ Identification of At-Risk Students using Random Forest:

Random Forest classification was applied to identify students who are academically at risk. Random Forest is an ensemble machine learning technique that combines multiple decision trees to improve prediction accuracy and reduce overfitting. It is highly effective for classification problems involving complex relationships among variables.

Criteria for At-Risk Classification:

Students were classified as "At-Risk" based on the following conditions:

- Attendance less than 60%
- Previous Percentage less than 50%
- Presence of Backlogs

Binary Variable Definition:

- 0 = Not at Risk
- 1 = At Risk

Model Specification:

- **Dependent Variable:** At-Risk Status (0/1)
- **Independent Variables:** Study Hours, Attendance, Motivation Score, Stress Level, Sleep Hours, Family Income, Internet Access, Parent Support, Internal Performance

Purpose of the Model: The Random Forest model helps in accurately identifying students who are at academic risk by capturing complex nonlinear relationships among multiple factors. This enables educational institutions to implement timely interventions and support mechanisms.

Results:

Model Performance Evaluation:

Confusion Matrix:

	Predicted: Not At-Risk (0)	Predicted: At-Risk (1)
Actual: Not At-Risk (0)	46	**1**
Actual: At-Risk (1)	**6**	**4**

Classification Report:

Class	Precision	Recall	F1-Score	Support
Not At-Risk (0)	0.88	0.98	0.93	**47**
At-Risk (1)	**0.80**	**0.40**	**0.53**	**10**

Overall Model Accuracy: 0.8772 (87.72%)

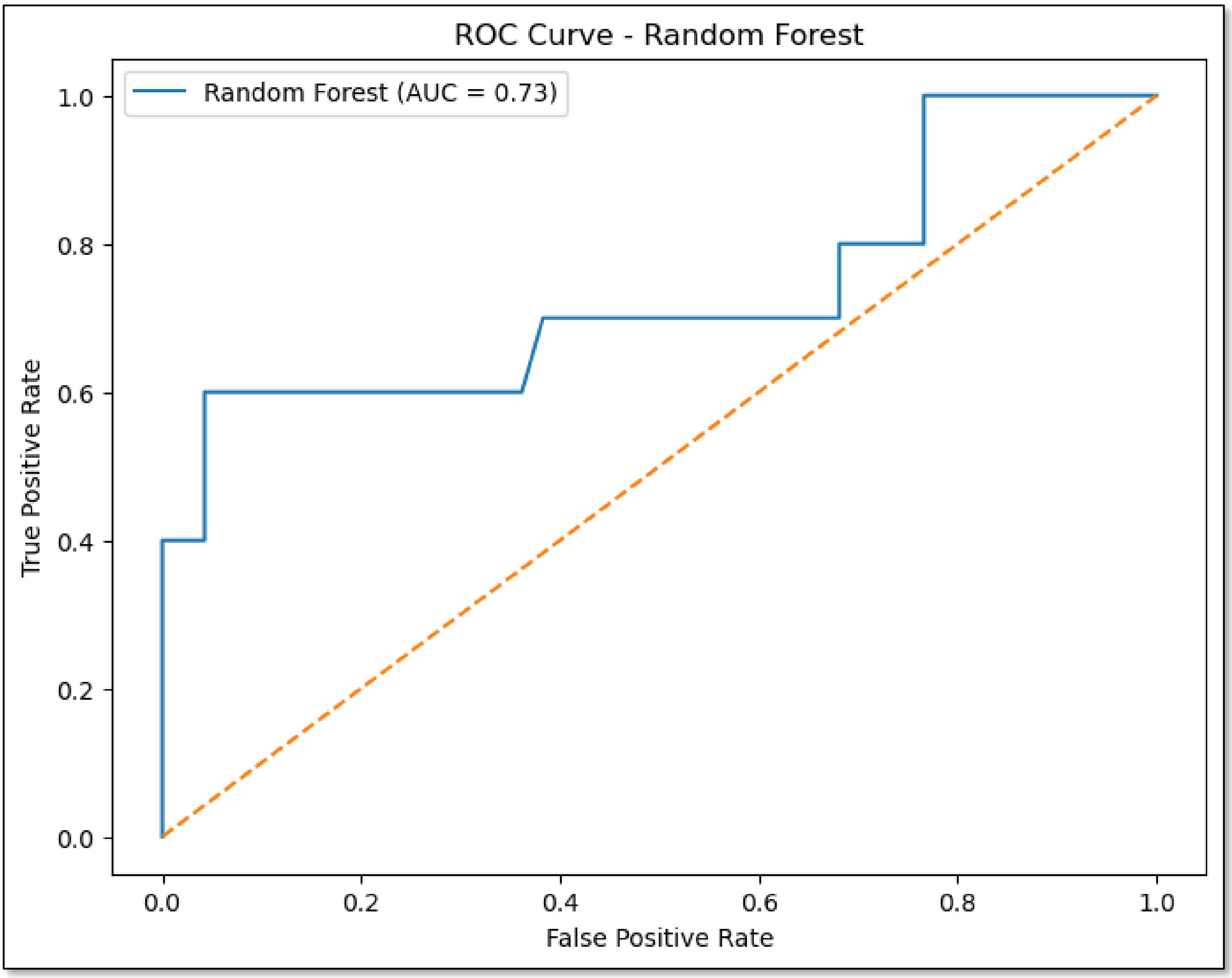
ROC Curve and AUC Analysis:

ROC Curve (Receiver Operating Characteristic Curve):

The ROC curve was plotted to evaluate the performance of the Random Forest model in distinguishing between at-risk and not at-risk students.

Area Under the Curve (AUC):

The AUC value obtained for the Random Forest model is **0.73**.



Interpretation:

Random Forest was used to identify academically at-risk students.
a) The model achieved an accuracy of 87.72%, indicating good overall performance.
b) Most non-risk students are correctly classified with very few errors.
c) For at-risk students, recall is 0.40 and precision is 0.80, showing moderate detection ability.
d) The F1-score (0.53) and AUC (0.73) indicate satisfactory performance and better discrimination ability.

The model performs well for non-risk students and shows moderate effectiveness in identifying at-risk students, with better overall classification compared to Logistic Regression.

➢ Identification of At-Risk Students using XGBoost:

XGBoost (Extreme Gradient Boosting) was applied to identify students who are academically at risk. XGBoost is a powerful ensemble machine learning algorithm based on gradient boosting, known for its high predictive accuracy, efficiency, and ability to handle complex data patterns.

Criteria for At-Risk Classification:

Students were classified as "At-Risk" based on the following conditions:

- Attendance less than 60%
- Previous Percentage less than 50%
- Presence of Backlogs

Binary Variable Definition:

- 0 = Not at Risk
- 1 = At Risk

Model Specification:

- **Dependent Variable:** At-Risk Status (0/1)
- **Independent Variables:** Study Hours, Attendance, Motivation Score, Stress Level, Sleep Hours, Family Income, Internet Access, Parent Support, Internal Performance

Purpose of the Model:

This model leverages advanced boosting techniques to identify academically at-risk students with high predictive efficiency and strong discriminative power.

Results:

Model Performance Evaluation:

Confusion Matrix:

	Predicted: Not At-Risk (0)	Predicted: At-Risk (1)
Actual: Not At-Risk (0)	45	**2**
Actual: At-Risk (1)	**6**	**4**

Classification Report:

Class	Precision	Recall	F1-Score	Support
Not At-Risk (0)	0.88	0.96	0.92	**47**
At-Risk (1)	**0.67**	**0.40**	**0.50**	**10**

Overall Model Accuracy: 0.8596 (85.96%)

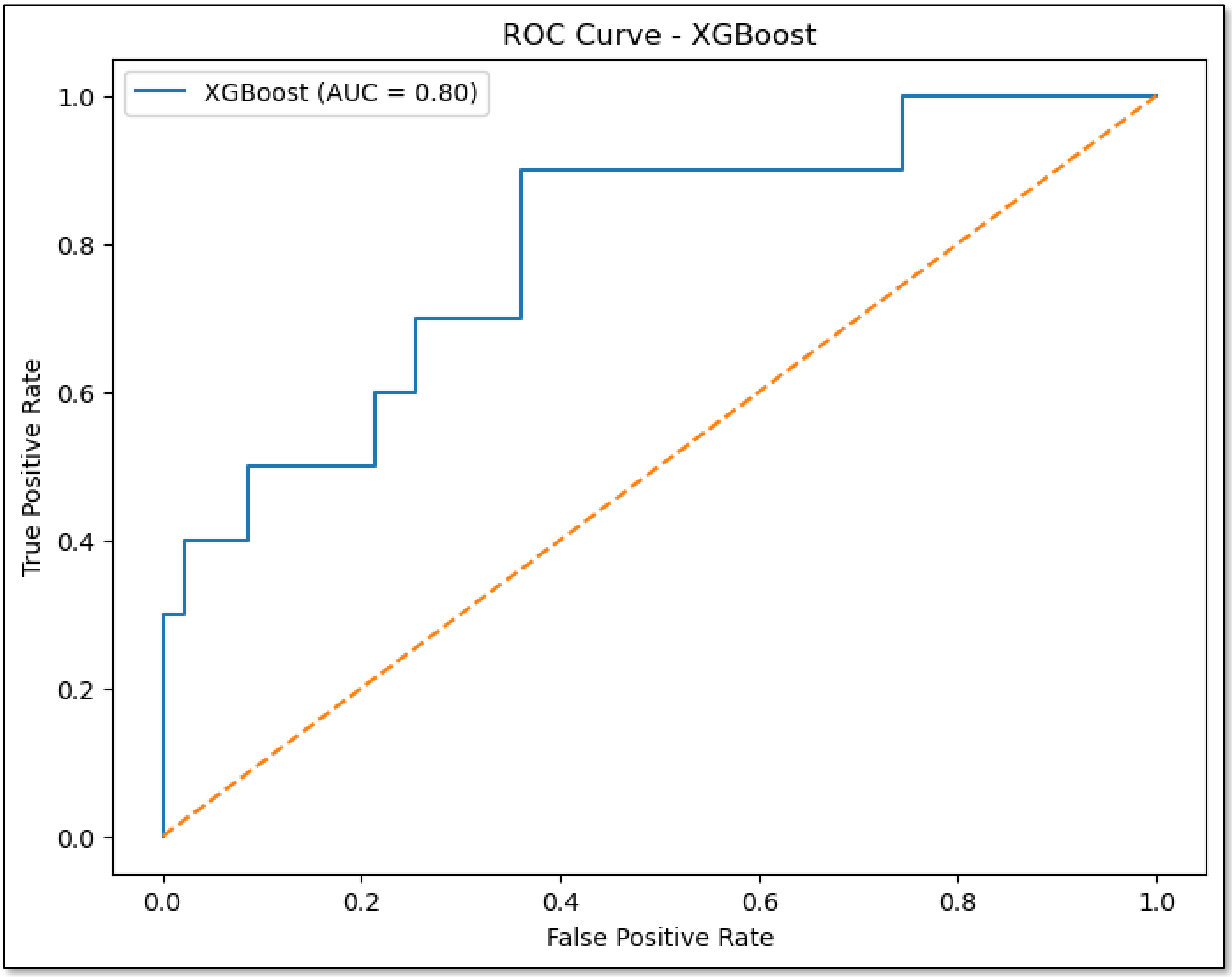
ROC Curve and AUC Analysis:

ROC Curve (Receiver Operating Characteristic Curve):

The ROC curve was plotted to evaluate the performance of the XGBoost model in distinguishing between at-risk and not at-risk students.

Area Under the Curve (AUC):

The AUC value obtained for the XGBoost model is **0.80**.



Interpretation

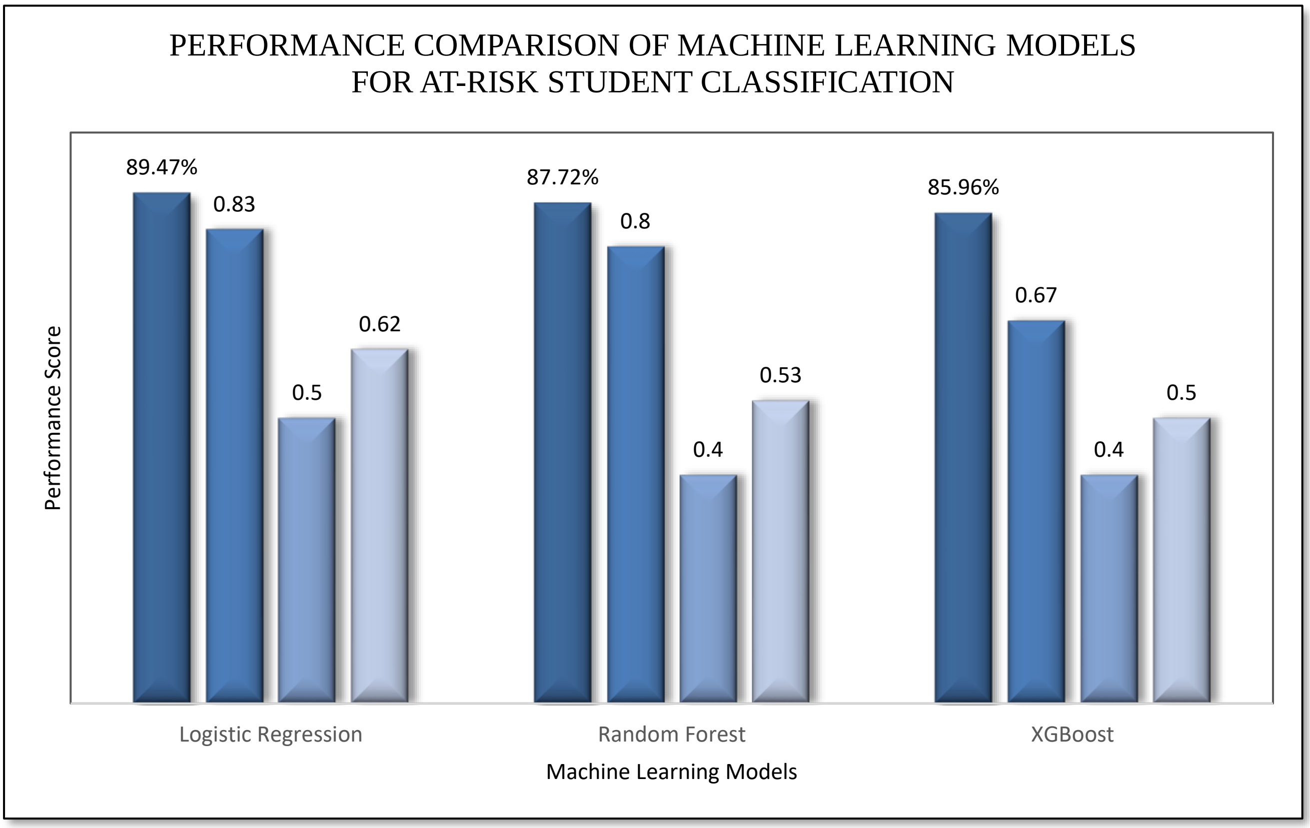
XGBoost was used to identify academically at-risk students.
a) The model achieved an accuracy of 85.96%, indicating good overall performance.
b) Most non-risk students are correctly classified with minimal errors.
c) For at-risk students, recall is 0.40 and precision is 0.67, showing moderate detection ability.
d) The F1-score (0.50) and AUC (0.80) indicate good classification and strong discrimination ability.

The model performs well for non-risk students and shows moderate effectiveness in identifying at-risk students, with the best AUC among all models.

➢ Performance Metrics for At-Risk Student Classification (Class = 1):

To evaluate the effectiveness of different machine learning algorithms in identifying academically at-risk students, three classification models—Logistic Regression, Random Forest, and XGBoost—were implemented and compared. The models were assessed using key performance metrics, including Accuracy, Precision, Recall, and F1-Score for the at-risk class (Class = 1). This comparative analysis helps determine the most suitable model for early identification and intervention.

Classifier	Accuracy	Precision (At-Risk)	Recall (At-Risk)	F1-Score (At-Risk)	Support (At-Risk N)
Logistic Regression	89.47%	0.83	0.5	0.62	**10**
Random Forest	87.72%	0.8	0.4	0.53	**10**
XGBoost	**85.96%**	**0.67**	**0.4**	**0.5**	**10**



Interpretation:

a) Logistic Regression shows the best overall performance with the highest accuracy (89.47%).
b) Random Forest also performs well with good accuracy (87.72%) and strong precision.
c) XGBoost gives slightly lower accuracy (85.96%) but has the best AUC.
d) All models show moderate ability in identifying at-risk students.

Overall, Logistic Regression is the best model, while Random Forest and XGBoost also give reliable results.

➢ Exploratory Factor Analysis (EFA) of Student Challenges:

Exploratory Factor Analysis (EFA) was applied to identify the underlying dimensions of student challenges and to group related variables into common factors.

Variables Included in Factor Analysis:

The following variables related to student behaviour, distractions, and support systems were included in the factor analysis:

- **Personal Factors:** Stress, Family Pressure, Sleep Hours
- **Study Behaviour:** Note Taking, Group Study, Self-Study, Online Tutorials, Practice Tests
- **Distractions:**Mobile Usage, Friends, Family, Lack of Interest, Social Media, Health Issues
- **Support Systems:**Extra Classes, Counselling, Study Materials, Time Management Support, Group Discussion, Career Guidance, Interactive Learning

Sampling Adequacy Test:

Test	Value
KMO (Kaiser-Meyer-Olkin)	0.555
Bartlett's Test (p-value)	< 0.001

Interpretations:

a) The KMO value of 0.555 indicates moderate sampling adequacy, suggesting that the data is acceptable for factor analysis.
b) The Bartlett's Test is statistically significant ($p < 0.001$), indicating that the variables are sufficiently correlated to proceed with factor analysis.

Factor Loading Matrix:

Variable	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5
Stress	-0.115	-0.137	-0.013	0.027	**0.031**
family_pressure	0.247	0.067	0.143	-0.152	**-0.02**
Sleep_Hours	0.085	0.149	0.07	0.113	**0.096**
study_note_taking	-0.106	0.218	0.106	-0.068	**-0.586**
study_group_study	**0.726**	0.045	0.163	-0.048	**-0.149**
study_self_study	-0.199	0.097	0.037	-0.11	0.71
study_online_tutorials	0.156	**0.523**	0.164	-0.06	**-0.012**
study_practice_tests	0.14	0.284	0.126	-0.048	**-0.136**
dis_mobile	-0.173	0.104	0.082	**0.84**	**-0.005**
dis_friends	0.219	-0.002	0.313	0.115	**0.088**
dis_family	0.161	0.049	0.38	-0.094	**-0.067**
dis_no_interest	0.099	0.165	**0.497**	0.003	**-0.045**
dis_social_media	0.004	0.286	0.204	-0.365	**0.072**
dis_health	-0.077	0.101	**0.569**	-0.015	**-0.022**
sup_extra_classes	**0.458**	0.155	-0.002	0.081	**0.089**
sup_counselling	-0.017	-0.083	-0.025	0.015	**0.042**
sup_materials	0.388	0.205	0.017	-0.182	**-0.06**
sup_time_mgmt	-0.038	0.288	0.197	0.048	**-0.001**
sup_group_discussion	0.372	0.389	0.103	-0.047	**-0.008**
sup_career	0.111	0.331	0.015	0.027	**0.029**
sup_interactive	**0.03**	0.559	**-0.029**	**0.01**	**0.006**

Interpretations:

Factor analysis grouped student challenges into key factors.

a) Factor 1 – Academic support and group study (e.g., group study, extra classes).
b) Factor 2 – Digital learning (e.g., online tutorials).
c) Factor 3 – Personal and health issues
d) Factor 4 – Distractions (mainly mobile)
e) Factor 5 – Self-study

Overall, student challenges are divided into main areas like study support, digital learning, personal issues, distractions, and self-study.

➢ Support Needs Analysis of Students:

To provide practical recommendations, Support Needs Analysis was conducted. Frequency analysis was performed to identify the most commonly required academic and personal support among students. Cross-tabulation was used to examine the relationship between challenge level and different types of support needs. Chi-square test was applied to determine whether there is a significant association between students' challenge levels and their support requirements. This analysis helps in identifying the types of support required by students facing higher levels of challenges and assists in designing targeted interventions.

Distribution of Support Needs among Students:

Support Type	Frequency
Time Management	**113**
Counselling	**98**
Career Guidance	**89**
Study Materials	**75**
Group Discussion	**31**
Extra Classes	**30**
Interactive Learning	**29**

Cross-tabulation Summary of Support Needs and Challenge Level:

Support Type	p-value	Result
Extra Classes	0.275	**Not Significant**
Counselling	0.65	**Not Significant**
Study Materials	0.548	**Not Significant**
Time Management	0.342	**Not Significant**
Group Discussion	0.193	**Not Significant**
Career Guidance	0.92	**Not Significant**
Interactive Learning	0.016	Significant

Interpretations:

Support Needs Analysis was conducted to identify required student support.
a) Time management (113) is the most needed support, followed by counselling (98) and career guidance (89).
b) Students with higher challenges tend to require more support, especially counselling and interactive learning.
c) Most support types do not show a significant association with challenge level ($p > 0.05$).
d) Interactive learning support shows a significant association ($p = 0.016$).

While general support is needed by all students, interactive learning is especially important for those facing higher challenges.

Major Findings

The present study examined the various academic, personal, and environmental factors affecting students' academic performance and identified the major challenges faced by undergraduate students. Based on the analysis, the following major findings were observed:

1. Students show moderate to good academic performance overall.
2. Academic performance is influenced by multiple factors such as study habits, attendance, motivation, stress, and family support.
3. Internal performance is the strongest positive factor, while backlogs negatively affect performance.
4. Study habits, attendance, and teacher support improve performance, whereas stress negatively impacts it.
5. Students face a moderate level of challenges, with higher challenges among Commerce and rural students.
6. Family income has no significant impact on academic performance.
7. Student challenges are grouped into key areas like academic support, personal issues, distractions, and self-study.
8. Time management and counselling are the most needed supports, while interactive learning is important for high-challenge students.
9. Machine learning models effectively identify at-risk students, with Logistic Regression performing best overall.

Recommendations

Based on the findings of the study, the following recommendations are suggested to improve students' academic performance and help them overcome academic and personal challenges:

1. Regularly monitor internal performance and give support to at-risk students through remedial classes and counselling.
2. Conduct time management, study skills, and motivation workshops to improve academic habits.
3. Strengthen counselling services to help students manage stress and personal challenges.
4. Use interactive and student-centred teaching methods for better learning.
5. Encourage teacher–student interaction and provide continuous academic guidance.
6. Provide special support to rural and weaker students who face more challenges.
7. Promote career guidance, peer learning, and group study to improve engagement.
8. Use machine learning models for early identification of at-risk students.

Overall, a balanced approach combining academic support, personal guidance, institutional care, and modern predictive analytics is essential for improving student success and overall well-being.

Conclusion

The present study examined the various academic, personal, and environmental factors affecting students' academic achievement and identified the major challenges faced by undergraduate students. The findings revealed that academic performance is influenced by multiple factors rather than a single variable. Among all the variables considered, internal assessment performance emerged as the strongest predictor of overall academic success.

The study found that students commonly experience moderate levels of academic and personal challenges. Issues such as lack of motivation, stress, difficulty in understanding subjects, and poor time management were frequently reported. Rural students and certain academic streams were observed to face comparatively higher levels of challenges.

Further analysis confirmed that personal experiences, study habits, and academic behaviour are closely associated with academic achievement. While factors such as family income, internal performance, and backlog status showed significant influence, other variables demonstrated comparatively modest individual effects.

The application of machine learning models, including Logistic Regression, Random Forest, and XGBoost, significantly enhanced the study by accurately identifying academically at-risk students. Among these models, Logistic Regression demonstrated the highest overall predictive performance, while Random Forest and XGBoost also proved highly effective and reliable.

The study underscores the importance of early identification of at-risk students and the implementation of timely academic interventions. It also emphasizes the need for effective counselling, interactive teaching methods, career guidance, and time management support to improve student outcomes.

Overall, student success depends on a combination of academic effort, personal well-being, family support, institutional assistance, and data-driven intervention strategies. Therefore, a comprehensive, student-centred, and technology-enabled approach is essential for improving academic performance and helping students overcome the challenges they face in their educational journey.

Limitations of the Study

Geographical Limitation
The study is confined to selected undergraduate colleges in Kolhapur District. Therefore, the findings may not be generalized to students in other districts or states with different socio-economic and educational contexts.

Cross-Sectional Design
The research is based on data collected at a single point in time. Hence, it identifies associations between variables but cannot establish definite cause-and-effect relationships.

Self-Reported Data
The study relies on self-reported responses from students. There is a possibility of response bias, social desirability bias, or inaccurate reporting of academic performance and personal challenges.

Limited Scope of Variables
Although the study includes multiple academic, personal, and environmental factors, other potential influences such as personality traits, institutional policies, teaching quality, or peer academic competition were not included due to time and resource constraints.

Sampling Constraints
Even though stratified sampling with proportional allocation was used, complete randomisation across all colleges in the district was not practically possible, which may slightly affect representativeness

Measurement Limitation
Likert-scale responses measure perceived levels of stress, motivation, and challenges, which are subjective in nature and may vary based on individual interpretation.

Model Limitation
The regression models used in the study explain a portion of the variation in academic achievement; however, academic performance is influenced by multiple complex factors that may not be fully captured in a single statistical model.

Overall, these limitations should be considered while interpreting the findings of the study.

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Appendix:

✦ Questionnaire:

1. **Gender / लिंग :**
   a) Male / पुरुष   b) Female / स्त्री   c) Transgender / ट्रान्सजेंडर

2. **Area / क्षेत्र :**
   a) Rural / ग्रामीण   b) Urban / शहरी

3. **Faculty/ शाखा:**
   a) Arts / कला
   b) Science / विज्ञान
   c) Commerce / वाणिज्य
   d) Management / व्यवस्थापन

4. **College Location / महाविद्यालयाचे ठिकाण :**
   a) Rural / ग्रामीण   b) Urban / शहरी

5. **Father's education / वडिलांचे शिक्षण :**
   a) Primary / प्राथमिक
   b) Secondary / माध्यमिक
   c) Higher Secondary / उच्च माध्यमिक
   d) Graduate / पदवीधर
   e) Post Graduate / पदव्युत्तर

6. **Mother's education / आईचे शिक्षण :**
   a) Primary / प्राथमिक
   b) Secondary / माध्यमिक
   c) Higher Secondary / उच्च माध्यमिक
   d) Graduate / पदवीधर
   e) Post Graduate / पदव्युत्तर

7. **Father's occupation / वडिलांचा व्यवसाय :**
   a) Unemployed (not currently working) / बेरोजगार (सध्या नोकरी नाही)
   b) Farmer / शेतकरी
   c) Daily wage labourer / मजूर
   d) Government employee / शासकीय कर्मचारी
   e) Private sector job / खाजगी नोकरी
   f) Business / व्यवसाय

8. **Mother's occupation / आईचा व्यवसाय :**
   a) Unemployed (not currently working) / बेरोजगार (सध्या नोकरी नाही)
   b) Farmer / शेतकरी
   c) Daily wage labourer / मजूर

d) Government employee / शासकीय कर्मचारी

e) Private sector job / खाजगी नोकरी

f) Business / व्यवसाय

9. Family Annual Income / कुटुंबाचे वार्षिक उत्पन्न :

a) Less than ₹1,00,000 / ₹१,००,००० पेक्षा कमी

b) ₹1,00,000–₹4,00,000

c) ₹4,00,000–₹8,00,000

d) Above ₹8,00,000 / ₹८,००,००० पेक्षा जास्त

10. What is your average class attendance? / तुमची सरासरी उपस्थिती किती आहे?

a) Less than 60% / ६० % पेक्षा कमी

b) 60%–75%

c) Above 75% / ७५% पेक्षा जास्त

11. How often do you attend regular lectures? / तुम्ही नियमित व्याख्यानांना किती वेळा उपस्थित राहता?

a) Rarely (only before exams) / क्वचित (फक्त परीक्षेच्या आधी)

b) Sometimes / कधी कधी

c) Most of the time / बहुतेक वेळा

d) Almost every lecture / जवळजवळ प्रत्येक व्याख्यानाला

12. Do you regularly submit assignments and classwork? / तुम्ही नियमितपणे असाइनमेंट व वर्गकार्य जमा करता का?

a) Never / कधीच नाही

b) Rarely / क्वचित

c) Sometimes / कधी कधी

d) Always / नेहमी

13. How often do you get academic help from teachers after class? / तुम्हाला वर्गानंतर शिक्षकांकडून किती वेळा शैक्षणिक मदत मिळते?

a) Regularly / नियमित

b) Sometimes / कधी कधी

c) Rarely / क्वचित

d) Never / कधीच नाही

14. How many hours do you study daily? / तुम्ही दररोज किती तास अभ्यास करता?

a) Less than 1 hour / १ तासापेक्षा कमी

b) 1–2 hours / १–२ तास

c) 2–4 hours / २–४ तास

d) More than 4 hours / ४ तासांपेक्षा जास्त

15. How good are you at managing your study and other activities? / अभ्यास आणि इतर गोष्टी सांभाळण्यात तुम्ही किती प्रभावी आहात?

a) Not effective at all / अजिबात प्रभावी नाही

b) Slightly effective / थोडे प्रभावी

c) Moderately effective / मध्यम प्रभावी

d) Very effective / खूप प्रभावी

e) Extremely effective / अत्यंत प्रभावी

16. What study methods do you primarily use? /तुम्ही मुख्यतः कोणत्या अभ्यास पद्धती वापरता? (Tick all that apply / लागू असलेले सर्व चिन्हांकित करा)

a) Note-taking / नोंदी करणे

b) Group study / गट-अभ्यास

c) Self-study / स्वयंअभ्यास

d) Online tutorials / ऑनलाईन ट्युटोरियल्स

e) Practice tests / सराव चाचण्या

f) Other (specify) / इतर (नमूद करा): ________

17. What was your previous year percentage? / मागील वर्षाचे टक्केवारी किती होती?

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18. How would you rate your internal assessment performance?/ तुमची अंतर्गत मूल्यांकन (Internal Assessment) कामगिरी तुम्ही कशी मोजाल?

a) Poor/खराब

b) Average /सरासरी

c) Good /चांगली

d) Very good / खूप चांगली

e) Excellent/ उत्कृष्ट

____

19.How many backlogs did you have last semester? / मागील सत्रात तुम्हाला किती बॅकलॉग होते?

a) None / एकही नाही

b) 1–2

c) 3 or more / ३ किंवा अधिक

20.Do you have access to the internet for study purposes, and if yes, is it college internet or your own? /अभ्यासासाठी इंटरनेटची सुविधा उपलब्ध आहे का? असल्यास, ती कॉलेजकडून आहे की स्वतःची?

a) Yes, college internet /हो, कॉलेजचे इंटरनेट

b) Yes, personal (self) internet /हो, स्वतःचे (वैयक्तिक) इंटरनेट

c) Both college and personal /दोन्ही (कॉलेज व स्वतःचे)

d) Sometimes /कधी कधी

e) No /नाही

21.How often do you use online platforms for studying? / तुम्ही ऑनलाईन प्लॅटफॉर्म (जसे YouTube & Other apps) किती वेळा वापरता?

a) Daily / दररोज

b) Few times in a week / आठवड्यात काही वेळा

c) Rarely / क्वचित

d) Never / कधीच नाही

22.How often do other responsibilities (e.g., part-time job,Other) affect your study time? / इतर जबाबदाऱ्या (उदा. पार्ट-टाईम नोकरी, घरची कामे) अभ्यासाच्या वेळेत किती अडथळा आणतात?

a) Never / कधीच नाही

b) Rarely / क्वचित

c) Sometimes / कधी कधी

d) Often / अनेकदा

e) Always / नेहमी

23.How much do you face the following challenges while studying? / अभ्यास करताना तुम्हाला खालील अडचणी किती प्रमाणात येतात?

a) Lack of study motivation / अभ्यासाची कमी प्रेरणा

☐ Not at all / अजिबात नाही ☐ Slightly / थोडेसे ☐ Moderately / मध्यम ☐ Very much / खूप जास्त ☐ Extremely / अत्यंत

b) Family problems / कौटुंबिक समस्या

☐ Not at all / अजिबात नाही ☐ Slightly / थोडेसे ☐ Moderately / मध्यम ☐ Very much / खूप जास्त ☐ Extremely / अत्यंत

c) Lack of Concentration / मन एकाग्र न होणे

☐ Not at all / अजिबात नाही ☐ Slightly / थोडेसे ☐ Moderately / मध्यम ☐ Very much / खूप जास्त ☐ Extremely / अत्यंत

d) Health issues / आरोग्य समस्या

☐ Not at all / अजिबात नाही ☐ Slightly / थोडेसे ☐ Moderately / मध्यम ☐ Very much / खूप जास्त ☐ Extremely / अत्यंत

e) Difficulties in understanding subjects / विषय समजण्यात अडचणी

☐ Not at all / अजिबात नाही ☐ Slightly / थोडेसे ☐ Moderately / मध्यम ☐ Very much / खूप जास्त ☐ Extremely / अत्यंत

f) Not enough time for study / अभ्यासासाठी वेळ कमी

☐ Not at all / अजिबात नाही ☐ Slightly / थोडेसे ☐ Moderately / मध्यम ☐ Very much / खूप जास्त ☐ Extremely / अत्यंत

24.What usually distracts you while studying? (Tick all that apply) / अभ्यास करताना तुम्हाला नेहमी कोणत्या गोष्टी विचलित करतात? (लागू त्या सर्व निवडा)

a) Mobile / TV / Laptop / मोबाईल / टीव्ही / लॅपटॉप

b) Friends / मित्र

c) Family responsibilities / कौटुंबिक जबाबदाऱ्या

d) Lack of interest / रस नसणे

e) Social media (WhatsApp, Instagram, etc.) / सोशल मीडिया (व्हॉट्सअ‍ॅप, इन्स्टाग्राम इ.)

f) Health issues / आरोग्य समस्या

g) Other (please specify): ________ / इतर (कृपया नमूद करा): ________

25.Do your parents support your studies? / तुमचे पालक तुमच्या अभ्यासाला पाठिंबा देतात का?

a) Yes / हो

b) No / नाही

26.What keeps you motivated to study? (Tick all that apply) / तुम्हाला अभ्यासासाठी प्रेरित ठेवणाऱ्या गोष्टी कोणत्या? (लागू त्या सर्व निवडा)

a) Family support / कुटुंबाचा आधार

☐ Not at all / अजिबात नाही ☐ Slightly / थोडेसे ☐ Moderately / मध्यम ☐ Very much / खूप जास्त ☐ Extremely / अत्यंत

b) Personal goals / वैयक्तिक उद्दिष्टे

☐ Not at all / अजिबात नाही ☐ Slightly / थोडेसे ☐ Moderately / मध्यम ☐ Very much / खूप जास्त ☐ Extremely / अत्यंत

c) Interest in subjects / विषयांमध्ये रस

☐ Not at all / अजिबात नाही ☐ Slightly / थोडेसे ☐ Moderately / मध्यम ☐ Very much / खूप जास्त ☐ Extremely / अत्यंत

d) Support from teachers / शिक्षकांचा आधार

☐ Not at all / अजिबात नाही ☐ Slightly / थोडेसे ☐ Moderately / मध्यम ☐ Very much / खूप जास्त ☐ Extremely / अत्यंत

e) Friends/peer encouragement / मित्रांचा / सहकाऱ्यांचा प्रोत्साहन

☐ Not at all / अजिबात नाही ☐ Slightly / थोडेसे ☐ Moderately / मध्यम ☐ Very much / खूप जास्त ☐ Extremely / अत्यंत

f) Fear of failure / अपयशाची भीती

☐ Not at all / अजिबात नाही ☐ Slightly / थोडेसे ☐ Moderately / मध्यम ☐ Very much / खूप जास्त ☐ Extremely / अत्यंत

27. Do you feel stressed or anxious about your studies? / अभ्यासाबद्दल तुम्हाला ताण किंवा चिंतेचा अनुभव येतो का?

a) Neutral / तटस्थ

b) Sometimes / कधी कधी

c) Often / अनेकदा

d) Always / नेहमी

28.Do you feel family pressure to study well academically? / चांगले शैक्षणिक यश मिळवण्यासाठी तुम्हाला कुटुंबाकडून दडपण येते का?

a) Never / कधीच नाही

b) Sometimes / कधी कधी

c) Often / अनेकदा

d) Always / नेहमी

29.How many hours do you sleep daily? / तुम्ही दररोज किती तास झोपता?

a) Less than 5 hours / ५ तासांपेक्षा कमी

b) 5–6 hours / ५–६ तास

c) 6–8 hours / ६–८ तास

d) More than 8 hours / ८ तासांपेक्षा जास्त

30.What kind of support do you personally need to study better? (Tick all that apply) / चांगले शिकण्यासाठी तुम्हाला वैयक्तिक पातळीवर कोणत्या प्रकारचा आधार हवा आहे? (लागू त्या सर्व निवडा)

a) Extra classes / अतिरिक्त वर्ग

b) Personal counselling / वैयक्तिक समुपदेशन

c) Study materials / अभ्यास साहित्य

d) Time management tips / वेळेचे व्यवस्थापन टिप्स

e) Group study discussions / गट-अभ्यास चर्चा

f) Career guidance / करिअर मार्गदर्शन

g) More interactive teaching / अधिक संवादात्मक अध्यापन